

AI/ML Mini Project Submission Format

1. Cover Page

Project Title: Machine Learning Based AI Chatbot using Python

Student Name - Pallevani Varshini

College Name - JBIET

Branch & Year – CSE & 2nd Year

Email / Phone Number – pallevenivarshini@gmail.com

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2. Project Overview

Problem Statement

Many users require quick responses and assistance for common queries. Traditional systems require manual interaction and are time-consuming. This project aims to develop an AI chatbot that can understand user messages and generate intelligent responses automatically using Natural Language Processing and Machine Learning techniques.

Objective

The main objective of this project is to build an intelligent AI chatbot that can understand user queries, classify intents, and provide suitable responses through an interactive web interface.

Technologies Used

Example:

- Python
- NLTK
- scikit-learn
- Streamlit
- JSON
- TF-IDF Vectorization
- Logistic Regression
- Machine Learning
- Natural Language Processing (NLP)

3. Dataset Details

Dataset Name- Custom Intent Dataset

Dataset Source (Kaggle / Custom Dataset)- Custom Dataset Created using intents.json

Number of Records or Images - 25+ Training Patterns

Dataset Link - Custom JSON Dataset

4. Project Workflow

- Data Collection
- Data Preprocessing
- Text Vectorization using TF-IDF
- Model Training using Logistic Regression
- Intent Prediction
- Response Generation
- Streamlit Web Interface Development
- Testing and Output Generation

5. Model / Algorithm Used

TF-IDF Vectorization

TF-IDF is used to convert text data into numerical vectors so that the machine learning model can process user messages effectively.

Logistic Regression

Logistic Regression is used for intent classification. It predicts the category of the user query based on trained data patterns.

Why These Algorithms Were Selected

- Easy to implement
- Fast training and prediction
- Suitable for text classification tasks
- Good accuracy for beginner AI chatbot projects
- Efficient for small datasets

6. Implementation

- streamlit
- sklearn
- nltk
- pickle
- Json

Files Used in Project

- app.py
- chatbot.py
- train_model.py
- intents.json
- model.pkl
- Vectorizer.pkl

Implementation Steps

1. Created intents dataset using JSON
2. Trained TF-IDF vectorizer
3. Trained Logistic Regression model
4. Saved model using pickle
5. Built chatbot frontend using Streamlit
6. Added chat history feature
7. Generated responses based on ML predictions

Output Screenshots

- Chatbot UI
- Model Training Output
- Chat History Interface

7.Results

Accuracy

The chatbot successfully predicts user intents and generates appropriate responses with good accuracy for basic conversational queries.

Output Predictions

- Greeting Prediction
- Help Query Prediction

- Goodbye Prediction
- Thank You Prediction

Performance Analysis

The chatbot provides fast responses and successfully classifies user intents using Machine Learning techniques.

8. Challenges Faced

- Installing and configuring required Python libraries
- Training the machine learning model correctly
- Managing chatbot response prediction
- Building an interactive frontend using Streamlit
- Fixing errors related to model loading and UI display

9. Conclusion

Summarize: The project successfully developed an AI chatbot using Python, NLP, and Machine Learning concepts. The chatbot can classify user intents and generate suitable responses through an interactive web interface.

Future Improvements

- Add voice assistant feature
- Add real-time API integration
- Improve chatbot accuracy using Deep Learning
- Add multilingual support
- Deploy chatbot online

10. References

Documentation Links

- [Python Documentation](#)
- [Streamlit Documentation](#)
- [scikit-learn Documentation](#)

- NLTK Documentation

Learning Resources

- YouTube Tutorials
- Articles on NLP and Machine Learning
- Streamlit Official Website.

11. Submission Files

Students should upload:

1. Project Report (PDF)
2. Source Code ZIP File
3. Screenshots of Output
4. PPT Presentation (Optional)

File Naming Format

Varshini_MLChatbot_Project.pdf